

DKIM - DomainKeys Identified Mail

What is DKIM?

- **DomainKeys Identified Mail** is an email authentication method designed to detect email spoofing
- It allows the receiver to check that an email claimed to have come from a specific domain was indeed authorized by that domain
- **DKIM** uses digital signatures to validate email authenticity, based on asymmetric cryptography (public/private key pairs)

How DKIM Works

1. Email Sending Domain:

- Creates public/private key pair
- Publishes public key in DNS TXT record
- Keeps private key secure on mail server

2. When Sending Email:

- Mail server selects headers to sign
- Generates hash of selected content
- Encrypts hash with private key = signature
- Adds signature to email header as DKIM-Signature

DKIM DNS Record Format

`v=DKIM1; k=rsa; p=MIGfMA0GCSqGS Ib3DQEBA...`

Record Format:

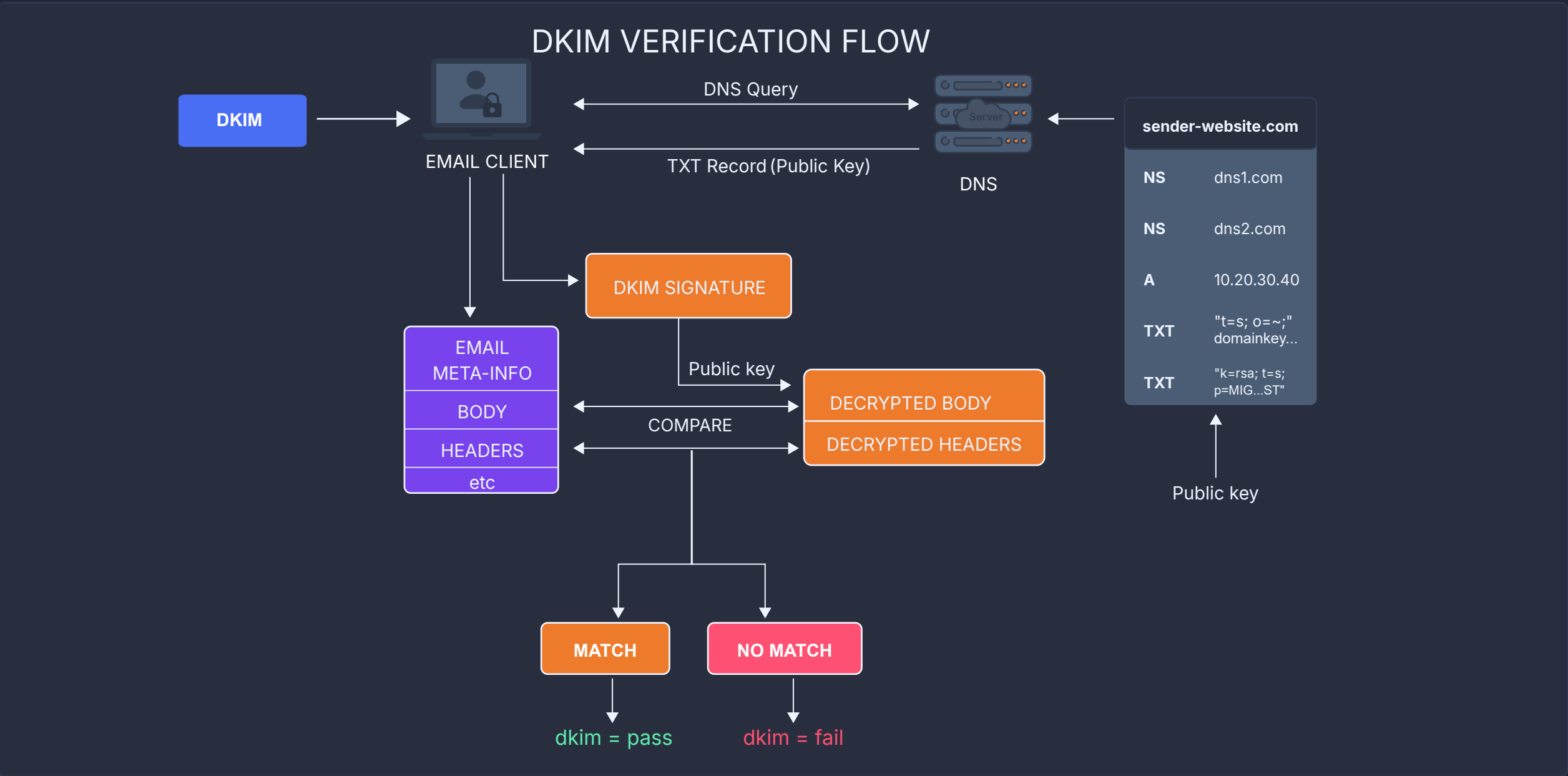
- **v=DKIM1**: Protocol version
- **k=rsa**: Key type (typically RSA)
- **p=...**: Public key (base64 encoded)
- **t=s**: Optional flags (s=strict, y=testing)
- **n=...**: Optional notes (human-readable)

DKIM DNS Setup

DNS TXT Record Location:

`selector._domainkey.example.com. IN TXT "v=DKIM1; k=rsa; p=MIGfMA0GCSqGS Ib3DQEBA..."`

- **selector**: A name that identifies this specific DKIM key (e.g., "mail", "2020", "dkim1")
- **_domainkey**: Fixed namespace for DKIM records
- **example.com**: Your domain name



DKIM Signature Header

`DKIM-Signature: v=1; a=rsa-sha256; d=example.com; s=selector1; c=relaxed/relaxed; q=dns/txt; t=1617998931; h=from:subject:date; bh=47DEQpj8HBSa+/TImW+5JCeuQeRkm5NMpJWZG3hSuFU=;`

- **v=1**: Version
- **a=rsa-sha256**: Algorithm
- **d=example.com**: Signing domain
- **s=selector1**: Selector
- **c=relaxed/relaxed**: Canonicalization
- **h=from:subject...**: Signed headers
- **bh=...**: Body hash